

Weighted Component Extraction and Persistent Elimination

Exact polynomial primitives towards
quasi-polynomial unknot recognition

Research continuation for Vladimir Reshetnikov’s ProveIt project
AI-assisted mathematical analysis, implementation, and independent audits

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Baseline: 9feb4346b4d050b305f825f0f7257f495960822a

Abstract

The ProveIt recognizer already contains exact compressed-word operations, binary interval-orbit counting, independent local orbit proofs, and a portfolio of knot simplification and decision procedures. Its remaining asymptotic obstacles are global: finding useful transformations, controlling the number of attempted transformations, and keeping every intermediate encoding small. This continuation develops two complementary improvements.

First, a weighted observer on the maintained interval proof computes all connected-component normal-coordinate vectors of a supplied surface, with binary multiplicities. This makes the classical weighted Agol–Hass–Thurston construction executable in the current code base, replaces dense many-port subset enumeration by sparse incidence, and extracts compressing disks from disconnected input surfaces. We prove a sharp three-run increase for the actual suffix-transfer schedule and an elementary support-sensitive bound: a surface with v triangulation vertices and q active quadrilateral coordinates has at most $v + 2q$ distinct component vectors, or $v + q$ when all components are two-sided.

Second, a signed persistent circuit represents an entire block of k raw singleton Tietze eliminations with $O(S_0 + k^2(D_0 + 1) + k^3 \log(D_0 + k + 2))$ vertices, where S_0 and D_0 are the initial circuit size and concatenation depth. A final export to an ordinary straight-line program at most doubles that size.

The accompanying code, tests, complete measurement samples, source hashes, and independent component audits make these claims reviewable. The results are polynomial primitives and a restricted representation theorem. They do not establish a general quasi-polynomial bound for unknot recognition. We state the missing composition hypotheses explicitly and propose a prioritized research programme for resolving them.

Reading guide. The main mathematical additions are the suffix-run bound, the component-profile theorem, and the whole-block elimination bound. The implementation sections explain exactly what a certificate proves. The measurements concern these supplied-input kernels; none is presented as a whole-knot speedup.

Contents

1	The research objective and the present result	3
1.1	Relation to the literature	3
1.2	Why this advances the asymptotic programme	4
2	Encodings, cost parameters, and proof boundaries	4
3	Weighted transport on a checked orbit trace	5
3.1	The exact full-range transfer	5
3.2	A sharp suffix-specific run bound	5
3.3	Sparse marked-component incidence	7
3.4	Connected normal-coordinate extraction	7
3.5	Exact scope of replay and limits	7
4	From component vectors to an exact disk witness	8
4.1	A bounded-size anchor construction	8
4.2	Euler characteristic without expanded disks	8
4.3	Essentiality on a torus must be tested per component	9
4.4	A mixed surface with enormous multiplicities	9
5	A linear bound on distinct normal component vectors	9
5.1	A sharp one-tetrahedron example	11
5.2	Consequences for encoded output	11
6	Polynomial representation of raw singleton blocks	11
6.1	Signed circuits and mutable generator bindings	12
6.2	Position-free extraction of a singleton context	12
6.3	The global size theorem	13
6.4	Running time and ordinary SLP export	14
6.5	A controlled family with exponentially long raw contexts	14
6.6	Consequences and next questions	15
7	Implementation, independent checks, and measurements	15
7.1	What changed in the maintained code	15
7.2	Tests that exercise mathematical failure modes	16
7.3	Geometry timing protocol	16
7.4	Batched follow-up and compressed capacity	17
7.5	What the word measurements establish	18
8	Certificate contracts and integration design	18
8.1	The weighted proof interface	18
8.2	Input provenance and the meaning of a negative result	19
8.3	Resource and cancellation semantics	19
8.4	A concrete integration sequence	19
9	Cost composition and the next research obligations	20
9.1	A conditional theorem for the whole computation	20
9.2	Prioritized questions and concrete targets	21
A	Reproduction and artifact map	22
B	A compact review checklist for the mathematical claims	23
	References	24

1 The research objective and the present result

The input problem is to decide whether a classical one-component knot diagram represents the unknot. Write n for its number of crossings, using a conventional finite planar-diagram encoding. An asymptotic target of quasi-polynomial time means a bound of the form

$$\exp(O((\log(n+2))^c))$$

for a fixed constant c , uniformly over all inputs. The often discussed bound $n^{O(\log n)}$ is the case $c = 2$. Neither a fast experiment nor a polynomial operation on a compressed object establishes such a bound: one must also bound how the object is found, how many states are examined, and the bit size of every intermediate state.

The present work continues the maintained algorithm at a pinned source revision. The source baseline is the ProveIt commit identified on the title page [8]. Its `Topology/UnknotRecognition/fast` tree contains the maintained implementation, while the `synthesis` tree records the accumulated mathematical and experimental analysis. Relevant existing modules include `interval_orbits`, `interval_orbit_verify`, `interval_incidence`, the normal-surface geometry and parity modules, `compressed_words`, and `elimination_batch`.

There are two concrete bottlenecks worth separating. The geometric code could count components and obtain aggregate topology, but it did not return the full vectors of the actual connected components of a general supplied disconnected surface. Its many-port incidence interface returned a dense array with one entry for every subset of the ports. The word code could apply a checked acyclic elimination batch, but a polynomial estimate for one batch did not itself control the size after many serial batches. We address both representation problems directly.

Result	Status in this continuation	Scope of the claim
Weighted interval orbits	Implemented over the maintained checked trace	Classical AHT theorem; new repository implementation
Three-run suffix bound	Proved, with a legal sharpness example	Full rightmost-range transfer followed by truncation
Sparse port incidence	Implemented and compared with the dense interface	Attained supports; no attachment order
Normal component inventory	Implemented, with source-bound replay	Supplied finite triangulation and normal vector
$H \leq v + 2q$, or $v + q$ for two-sided components	Proved and independently audited	Distinct component vectors; no priority claim
Persistent singleton blocks	Proved and implemented as a reference prototype	Raw singleton eliminations, without intervening normalization
Quasi-polynomial bound for this recognizer	Unresolved	Surface/move discovery, growth, branching, and provenance remain obligations

Table 1: A precise ledger of what has and has not been established.

1.1 Relation to the literature

Weighted interval-orbit counting is established by Agol, Hass, and Thurston, whose Theorem 16 and Corollary 17 include polynomial computation of normal-component vectors and multiplicities [1]. We use that result as mathematical infrastructure. Our transfer proof below is written for the precise conventions and trace rules in the maintained code. The improved constant comes from its full rightmost-range schedule, not from a new solution of the general orbit problem.

Normal-coordinate uniqueness and vertex-link structure are standard [2]. The profile bound developed here is an elementary deduction from those facts and disjointness. It belongs in the context of classical Haken–Kneser finiteness: for example, King’s discussion records an older linear component bound and explicitly notes that stronger bounds were already known [5]. We make no claim that our support-sensitive formulation is new in the literature.

Compressed group words and polynomial free reduction are also established [9]. Recent work of Ennes and Maria relates compressed Heegaard data to short group presentations [3]. Our word contribution is an explicit cumulative size proof for the particular mutable-binding representation and a tested implementation. The proof does not rely on identifying this construction as unprecedented.

The global target must be described carefully. Oxford’s 2021 announcement states Lackenby’s quasi-polynomial unknot-recognition result [7]. The 2026 preprint on incompressible surfaces and hierarchies provides a concrete new topological route and discusses future acceleration [6]. Its Section 9 explicitly distinguishes a bound on iterative steps from a precise running-time analysis. A parameter linear in the initial combinatorial complexity must not be silently treated as logarithmic. We therefore use this work to identify geometric operations and proof obligations, without importing a quasi-polynomial bound for our implementation. The source version used here is arXiv:2607.23350v1, dated 25 July 2026.

1.2 Why this advances the asymptotic programme

The geometry improvement turns a large normal sum into a small list of actual connected pieces. It can reveal a useful disk that aggregate invariants obscure, without enumerating exponentially many parallel copies. Its sparse incidence output also removes an exponential dependence on the number of ports when the caller only needs attained signatures. These are precisely the forms of information that a compressed cutting or hierarchy procedure would need to reuse.

The word improvement removes a serial reconstruction mechanism from an entire restricted block. A linear number of eliminations can be represented with polynomial encoded size even when the raw words become exponentially long. This is stronger than merely demonstrating a small grammar for one supplied giant word. The proof applies to every valid raw singleton donor sequence in the stated model.

Both advances reduce local sources of growth. They deliberately leave the harder global question visible: why should a sufficiently short sequence of useful geometric or algebraic operations exist and be discoverable for every unknot? The final research agenda specifies what a proof would need to answer that question.

2 Encodings, cost parameters, and proof boundaries

A binary interval system consists of a universe $I_N = \{0, \dots, N - 1\}$ and k partially defined translations or reflections between integer intervals. Pairing endpoints use binary integers, so the relevant size is $k \log(N + 1)$, not N . The maintained pairing objects use inclusive endpoints; weight intervals in this continuation are half-open. An explicit distinction prevents off-by-one errors at reflections and truncations.

A finite normal surface in a triangulation with t tetrahedra has a standard vector in $\mathbb{N}^{7t}$: four triangle coordinates and three quadrilateral coordinates per tetrahedron. Nonnegativity, the matching equations, and the quadrilateral constraint describe an embedded normal surface. At most one quadrilateral coordinate per tetrahedron is nonzero. We write $q \leq t$ for the number of active quadrilateral coordinates, v for the number of manifold vertices, and D for the total number of normal disks. The coordinate input size is $O(t \log(D + 1))$ under the usual uniform binary bound; it is not D .

The theorem on component profiles is stated for finite triangulations of compact manifolds with manifold vertex links. The implemented adapter is more restrictive: its existing geometry validator accepts connected, compact, orientable finite triangulations with one torus boundary component. It rejects unsupported or invalid geometry before constructing the orbit system. Ideal triangulations, spun surfaces, singular vertex links, and general higher-genus boundary patterns are not implicitly included.

For words, S denotes allocated circuit vertices, M the fixed list of original relator slots, and r_0 the initial generator count. A length of 2^{10000} has a short binary encoding and may have a small circuit; a traversal that performs 2^{10000} letter operations is nevertheless unacceptable. Conversely, many successive small transformations can create a huge circuit even if each transformation is polynomial in its current input. The whole-block proof below controls that second problem under explicit hypotheses.

All polynomial assertions use exact integer arithmetic with bit costs. Hash tables are implementation conveniences rather than a proof of deterministic worst-case bounds. Tests against expanded data are performed only on small systems; the production interval and word operations do not use expansion as their large-input fallback. Resource exhaustion returns an inconclusive result rather than a false negative.

3 Weighted transport on a checked orbit trace

Let $I_N = \{0, \dots, N-1\}$ and let $\sim$ be the equivalence relation generated by the input interval isometries. The input weight function is $w : I_N \rightarrow \mathbb{Z}^d$, represented by an additive list of m half-open constant intervals. An endpoint sweep converts it into a canonical complete partition with at most $2m+1$ runs, including any zero-weight runs. For a component O the requested weight is

$$W(O) = \sum_{x \in O} w(x).$$

The output is a finite list of distinct pairs (z, μ_z) , where μ_z counts the components whose weight vector equals z . This represents all components without enumerating them. Zero vectors, negative coordinates, and $d=0$ are legitimate; zero weight never means that a component is absent.

This is an implementation of the weighted Agol–Hass–Thurston algorithm [1, Section 6], not a new general orbit-counting theorem. The existing scheduler supplies a complete local orbit proof. The ordinary independent verifier checks that proof, after which a weight observer follows the same operations. The observer does not make scheduling decisions. Identity deletion, reflection trimming, the sharp or classical periodic merger, and transmission preserve the actual equivalence relation on the current points, so they leave their weights unchanged.

3.1 The exact full-range transfer

Immediately before a truncation the carrier has range $[c, N)$ and domain $[a, b+1)$, with $a < c$. The entire range is transferred, including its part that will remain after truncation. This full transfer is essential for the small breakpoint bound; repeatedly transferring only removed pieces is not the operation analysed here.

For a positive carrier put $p = c - a$. A range point x is moved to

$$\rho(x) = a + ((x - a) \bmod p) \in [a, c).$$

This is obtained by iterating the carrier’s partial inverse until its image first lies before c . Every intermediate inverse is defined. If the domain and range are disjoint, only one inverse is needed, and the same formula applies.

Suppose w equals v on the range subinterval $[l, h)$. Write $h - l = qp + r$, $0 \leq r < p$. The pushforward of this interval adds qv on all of $[a, c)$ and one more copy of v on the cyclic interval of length r beginning at $a + ((l - a) \bmod p)$. A cyclic interval is represented by one ordinary interval if it does not wrap and by two if it does. The formula uses one integer division; it performs no loop of length q .

For a trimmed reversing carrier the range point x is moved once to $a + N - 1 - x$. A constant range interval $[l, h)$ therefore adds its value to $[a + N - h, a + N - l)$. In either case the old range weights are first set to zero. Only their transferred copies remain. Every original contribution is moved to exactly one point in the same orbit, proving preservation of every orbit sum, even for signed vector weights. The existing truncation rule then removes a zero-weight suffix whose points have earlier orbit representatives.

At static contraction, every removed point represents one complete orbit. Intersect each static gap with the current constant-weight runs and add its integer length to the multiplicity of that vector. Shift the remaining run endpoints by deleted rank, just as the ordinary interval points are shifted. This gives two executable invariants:

$$\sum_z \mu_z = C, \quad \sum_z \mu_z z = \sum_{x \in I_N} w(x),$$

where C is the independently verified orbit count. These invariants are necessary checks; they alone would not prove the individual component weights, which are justified by the local transport argument.

3.2 A sharp suffix-specific run bound

The general AHT transfer lemma permits an increase of four constant intervals. Because the maintained scheduler always transfers a full rightmost range, its bound improves to three.

Proposition 3.1 (Three-run suffix bound). *Suppose the current nonempty universe has m constant-weight runs. Transfer the full carrier range $[c, N)$ as above and truncate to $I_{N'}$, where $c \leq N' < N$. The resulting weight function has at most $m+3$ runs. If $N' = c$, or if $a = 0$, it has at most $m+2$ runs. The bound $m+3$ is attained by a legal step of the maintained scheduler.*

Proof. Let $J = \{j : 0 < j < N, w(j-1) \neq w(j)\}$, so $m = 1 + |J|$. Put $s = |J \cap (c, N)|$. Old changes strictly below c may remain. The s old changes inside the transferred range disappear there.

For a disjoint positive carrier or a trimmed reflection, a pushforward has changes only at the two source endpoints and at the images of those s old changes. Setting the surviving part of $[c, N)$ to zero may add a change at c . Thus there are at most $s + 3$ new potential changes, which replace the s removed changes.

For an overlapping positive carrier, regard the pushforward on $[a, c)$ as a cyclic residue sum. Its interior changes can occur only at residues of the s old changes and at the residue of N . The initial range endpoint c has residue a . Extending the pushforward by zero can add source-boundary changes at a and c . Hence all changes again belong to the old changes below c , the at most s folded old changes, and three additional points: a , c , and the residue of N . Coincident points, zero vector changes and pre-existing changes only decrease their number. A source endpoint at zero is not an interior change. Nor is c an interior change when $N' = c$, giving the stated improvements.

For sharpness, let $N = 7$, use constant weight 1, and take the three pairings

$$[1, 2] \longrightarrow [5, 6], \quad [3, 4] \longrightarrow [4, 5], \quad [0, 0] \longrightarrow [3, 3].$$

All points are initially covered. The first pairing is the rightmost carrier. There is no periodic merger and neither remaining range lies inside its range. The second pairing forces the truncation point $N' = 6$. Full-range transfer produces values 1, 2, 1, 0 respectively on $[0, 1)$, $[1, 3)$, $[3, 5)$, $[5, 6)$. Thus one run becomes four. $\square$

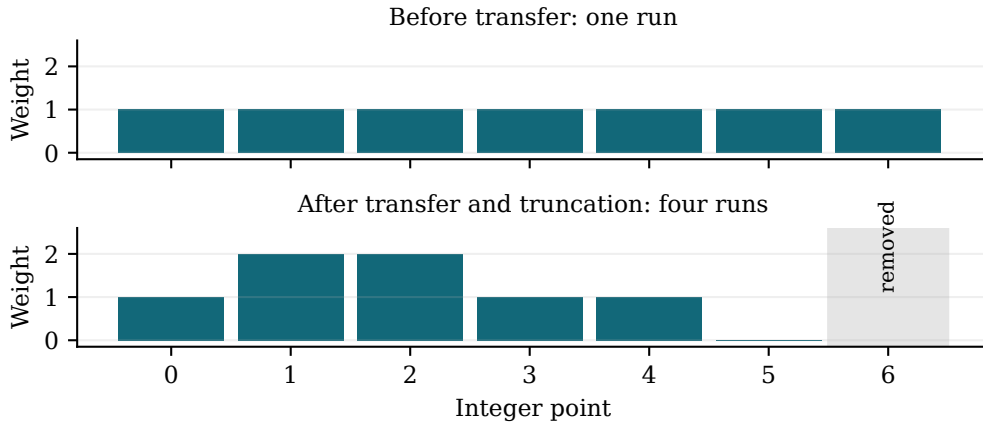


Figure 1: The exact seven-point example in the proof. Transferring the carrier range $[5, 7)$ to $[1, 3)$ and removing point 6 increases the number of constant runs from one to four. Zero weight at surviving point 5 remains part of the representation.

Static contraction cannot increase the number of weight runs: it merely takes an ordered subsequence of point values and coalesces equal adjacent runs. All other operations leave weights fixed. Consequently, after T full transfers the current weight partition has at most $m_0 + 3T$ runs. This bound is independent of the magnitudes of all represented intervals and multiplicities. It remains valid with the Fine–Wilf merger because that merger does not alter point weights or the equivalence relation.

For signed input weights let $A = \sum_j (h_j - l_j) \|v_j\|_1$, using the additive input intervals. Every intermediate coordinate has absolute value at most A : weights are gathered, never copied into two different orbit representatives. The bit length is therefore $O(1 + \log(A + 1))$, bounded by $O(\log(N + 1) + \log(m + 1) + L + \log(d + 1))$ when input coordinate bit length is at most L . The degree-zero case is treated separately with empty vectors. With the existing polynomial-size orbit trace, these facts give polynomial bit complexity for canonicalization, all vector arithmetic, and the compressed output. No new polynomial degree for the scheduler is asserted. A simple bound on the number of emitted distinct records is the number of trace contractions times $m_0 + 3T$, still polynomial.

3.3 Sparse marked-component incidence

For each of r ports A_i represented as a union of intervals, assign coordinate i of $w(x)$ to be 1 if $x \in A_i$ and 0 otherwise. Normalize the intervals within each port so overlaps do not accidentally count a point twice. The weighted output gives

$$W_i(O) = |O \cap A_i|,$$

which contains both exact point multiplicity and the incidence signature $\{i : W_i(O) > 0\}$. Merge output vectors having the same support to obtain a sparse histogram of attained signatures. Empty support is retained. With M marked intervals, this computation has polynomial bit complexity in $k, M, r, \log(N+1)$; there is no 2^r output requirement. If the caller requests a dense array indexed by every subset of $[r]$, that conversion still requires $\Omega(2^r)$ slots. This sparse interface is an application of the classical weighted theorem, not a new general result about AHT.

Additive incidence does not determine attachment order. A single circle with four distinctly labelled marked points in cyclic order a, b, c, d has the same weight vector $(1, 1, 1, 1)$ as one with order a, c, b, d . These orders are inequivalent under rotation and reversal with the labels fixed. Thus even exact multiplicities and supports cannot recover the ordered boundary data needed by a general cutting or gluing procedure. Such a procedure must retain an additional order and transport structure.

3.4 Connected normal-coordinate extraction

For each of the at most $7t$ normal-disc types choose one incident local edge. Add the corresponding basis vector at precisely the edge points met by that disc stack on that chosen local edge. A triangle at vertex v occupies the points nearest v . On an edge directed a to b , an incident quadrilateral stack occupies the interval after the a -triangle stack and before the b -triangle stack. The actual global edge orientation may reverse the interval; obtain it from the validated parity data. Different local edges may represent one global edge, so these basis contributions are additive and must not overwrite each other.

Each local normal disc contributes one basis unit exactly once. Therefore the orbit weight is the complete normal-coordinate vector of that actual connected component. Grouped multiplicities encode equal components. Their coordinate sum with multiplicity recovers the supplied vector. This construction works for one-sided components directly and needs no common-divisor reduction. In particular, 2 times a one-sided component may describe a connected two-sided double instead of two components; the orbit extraction, rather than linear multiplicity guessing, decides which occurs. This is the classical AHT Corollary 17 construction made executable over the maintained edge-arc encoding.

Once weighted extraction has proved connectedness, Euler characteristic one and nonempty boundary suffice to identify a disc, as proved in Lemma 4.1. The existing torus-boundary homology test then decides whether its boundary is essential. No further boundary or orientation orbit query is needed. Extracting such a vector from a disconnected input is new functionality of this code base. It does not provide surface search, diagram-to-exterior provenance, hierarchy construction, or a quasi-polynomial complete recognition procedure.

3.5 Exact scope of replay and limits

The weight certificate binds the normalized weight function and dimension, the source-bound unweighted trace, and the ordered exact output classes. Replay invokes no orbit scheduler. The ordinary orbit checker is independent of orbit search. The weight producer and weight checker share the small transport arithmetic; its independent validation comes from the expanded graph and direct pushforward tests. Calling the entire weight verifier arithmetically independent of weight production would overstate this boundary.

The producer's cycle limit applies only to ordinary orbit search. The weight-run, output-record and event limits are checked in the subsequent weighted replay; the full ordinary trace has already been generated and validated by then. They are cooperative work/output limits, not memory or wall-clock guarantees. The standalone replay and weight-certificate checker reject an over-limit trace by its explicit operation count before ordinary proof replay. Every incomplete producer or standalone replay returns no orbit count, class list, aggregate or certificate. Cancellation exceptions propagate.

4 From component vectors to an exact disk witness

4.1 A bounded-size anchor construction

The arc encoding uses one point for each intersection of the surface with a global triangulation edge. Normal arcs in faces pair appropriate intervals of those points. Around a local normal disk the boundary arcs connect all its corners, and gluing across faces identifies the same surface. Consequently the connected components of this interval graph are exactly the connected components of the normal surface.

For every active local disk type, choose one incident local edge and put its coordinate basis weight at the corresponding stack of edge points. Each disk contributes exactly once. The implementation explicitly maps the chosen edge interval into the canonical global edge orientation; identified local edges add weights at the same global points. The weight dimension is $7t$, and the number of additive anchor intervals is at most

$$h \leq 4t + q \leq 5t.$$

Canonicalizing their endpoints therefore gives at most $2h + 1 \leq 10t + 1$ initial weight runs. The face-arc construction uses at most $12t$ pairings, while $N \leq 4D$. The resulting weighted input is polynomial in t and $\log(D + 1)$.

The output vector of an orbit has a stronger meaning than a compatible summand of the input vector: it is the vector of one actual connected component. This distinction matters because normal coordinate addition can involve exchanges and reconnect summands. Verifying only that proposed vectors are nonnegative, satisfy matching, and sum to the source would not certify a component decomposition. The orbit trace supplies the missing connectivity information.

The adapter revalidates each extracted vector against the finite geometry and computes its Euler characteristic and boundary normal-arc count. It also checks the exact conservation identity

$$\sum_{j=1}^H \mu_j f_j = f,$$

where f is the supplied vector and μ_j are binary multiplicities. This is an additional consistency check, not a replacement for weighted connectivity replay.

4.2 Euler characteristic without expanded disks

Let T and Q be the sums of the triangle and quadrilateral coordinates of an extracted component. Its number of normal disks is $T + Q$, and its number of local normal-arc occurrences is $A = 3T + 4Q$. If B is the number of normal arcs on unglued boundary faces, its surface cell structure has $(A + B)/2$ edges: interior arcs are counted twice in A , while boundary arcs are counted once. Its vertices are the distinct intersections with global edges, counted by N_f . Thus

$$\chi(f) = N_f - \frac{A + B}{2} + T + Q. \quad (1)$$

Every quantity is obtained by linear arithmetic on binary counts. No loop over disks, arcs, or vertices is needed. The manifold and matching checks ensure that these are the cells of the embedded surface being described.

Lemma 4.1 (Connected disk shortcut). *Let F be a connected compact surface. If $\chi(F) = 1$ and $\partial F \neq \emptyset$, then F is a disk.*

Proof. For an orientable connected surface with genus g and $b \geq 1$ boundary components, $\chi = 2 - 2g - b$. Equality to one forces $g = 0$ and $b = 1$. For a nonorientable connected surface with $h \geq 1$ crosscaps and $b \geq 1$, $\chi = 2 - h - b \leq 0$, so equality to one is impossible. $\square$

The weighted orbit result already certifies connectedness. Consequently the test $\chi(f_j) = 1$ and $B_j > 0$ suffices to recognize a disk component; there is no need to run an orientation-double orbit computation or count its boundary components separately. A closed projective plane has Euler characteristic one but fails the boundary test, exactly as required. This is a computational shortcut derived from information already proved, not a change to surface classification.

4.3 Essentiality on a torus must be tested per component

A disk in a manifold is a compressing disk for its boundary if its boundary curve is essential in that boundary. On a torus an embedded simple closed curve is essential exactly when its mod-two homology class is nonzero. Indeed, an essential curve has a primitive integral slope (p, q) ; the coordinates cannot both be even. Conversely, a contractible curve is homologically zero. This uses the fact that the curve is a single embedded simple closed curve, which the disk shortcut has established.

The existing finite boundary parity module constructs and independently checks the corresponding cohomology witness. The new adapter invokes that module only on extracted disk profiles. It never substitutes an aggregate boundary class for the class of an individual disk. Two parallel meridians have zero total class modulo two, although each bounds a compressing disk. Such cancellation is one reason why component extraction adds information that aggregate topology cannot provide.

For a boundary component of higher genus, an essential separating curve may be homologically zero. The torus test would then be insufficient. Extending the inventory to such manifolds is plausible, but extending its essentiality conclusion requires a new boundary-curve decision procedure. The current validator's torus hypothesis is part of the theorem used by the implementation.

4.4 A mixed surface with enormous multiplicities

Take an eight-tetrahedron layered solid torus, a normal meridian disk with vector d , and its boundary vertex-link disk with vector ℓ . For $K = 2^b$, supply

$$f = Kd + (K + 1)\ell. \quad (2)$$

The vertex link can be taken in a sufficiently small vertex neighbourhood, disjoint from the meridian. The components are K meridian disks and $K + 1$ vertex-link disks. Their multiplicities are coprime. A global greatest-common-divisor extraction cannot explain this decomposition.

The inventory returns two profiles, with total component count $2K + 1$, and identifies exactly K compressing disks. The benchmark includes $b = 128, 1024, 4096, 16384$; the final case represents $2^{16385} + 1$ connected components. It completes by transporting binary weights on the same small combinatorial orbit trace. The certificate is not constant size: its repeated endpoints and multiplicities must still carry $O(b)$ bits. The complete serialized certificates and their sizes are included in the results.

One-sided surfaces give a different reason to avoid multiplicity guessing. A normal Möbius band with vector m has a connected annular frontier with vector $2m$. An odd multiple can contain both a Möbius component and annular components. The extraction follows actual connectivity, so it handles this directly. The next section proves that the number of distinct profiles nevertheless remains linearly bounded by the triangulation and the active quadrilateral support.

5 A linear bound on distinct normal component vectors

Weighted orbit extraction produces a compressed inventory even when the number of connected components is enormous. For an actual embedded normal surface, its fully coalesced inventory satisfies a stronger, linear bound. We prove it from normal-coordinate uniqueness and elementary normal neighborhoods. This is a finiteness refinement for the present analysis; we make no claim of literature-wide priority. Classical normal-surface finiteness is discussed by King [5, Theorem 2], who attributes the underlying result to Haken and Kneser and notes that stronger bounds were already known.

Let $\mathcal{T}$ be a finite triangulation of a compact 3-manifold M , possibly with boundary. Tetrahedral faces may be identified within or between tetrahedra, but the quotient must be a manifold: every vertex link is a connected sphere or disc. Ideal vertices, singular vertex links, and immersed surfaces are outside this statement. Write t for the number of tetrahedra and v for the number of quotient vertices. Orientability of M is not required.

For a properly embedded normal surface F , its standard vector $\mathbf{x}(F) \in \mathbb{Z}_{\geq 0}^{7t}$ records four triangle counts and three quadrilateral counts in each tetrahedron. These counts satisfy the standard matching equations and the quadrilateral constraints. Standard vectors determine normal surfaces up to ambient normal isotopy, including their connected component multisets [2, Lemma 2.5 and Theorem 2.10]. Let $Q(F)$ be the set of nonzero quadrilateral coordinates and put $q = |Q(F)| \leq t$. Count a component vector only once, regardless of how many components have that vector, and denote the resulting number by H .

Proposition 5.1 (Component-type bound). *Under these hypotheses,*

$$H \leq v + 2q.$$

If every component of F is two-sided, then $H \leq v + q$. Neither conclusion assumes incompressibility, least weight, or fundamentality of F .

Lemma 5.2 (Independence of disjoint two-sided types). *Mutually disjoint, connected, two-sided normal surfaces with pairwise distinct standard vectors have linearly independent standard vectors.*

Proof. Write the vectors as $\mathbf{y}_1, \dots, \mathbf{y}_h$. Since their entries are integers, a nontrivial real linear dependence implies a nontrivial rational dependence: row reduction of the corresponding integer matrix computes its kernel over $\mathbb{Q}$. Clearing denominators gives an integer dependence. The vectors are nonzero and coordinatewise nonnegative, so the nonzero coefficients cannot all have the same sign. Consequently a dependence would give

$$\sum_{i \in P} a_i \mathbf{y}_i = \sum_{j \in R} b_j \mathbf{y}_j, \quad a_i, b_j \in \mathbb{Z}_{>0}, \quad P \cap R = \emptyset,$$

with both index sets nonempty.

Choose mutually disjoint, sufficiently small normal product neighborhoods of the surfaces. Two-sidedness permits any positive number of parallel normal copies inside each such neighborhood, each with the same standard vector as the original. The two sums therefore represent actual disjoint unions having precisely the displayed component multisets. Equality of their full vectors gives a normal isotopy between the unions. Such an isotopy sends each connected component to a connected component and preserves its normal-disc counts. Thus some $\mathbf{y}_i$ on the left equals a $\mathbf{y}_j$ on the right, contradicting distinctness. $\square$

Proof of Proposition 5.1. Let $V_Q \subseteq \mathbb{R}^{7t}$ be the real solution space of the standard matching equations with all quadrilateral coordinates outside $Q(F)$ set to zero. Project V_Q to its q permitted quadrilateral coordinates. Its kernel has dimension exactly v .

Indeed, when every quadrilateral coordinate vanishes, face matching identifies the triangle coefficients at the corners joined through that face. The corner triangles around each quotient vertex have connected adjacency graph, because its link is a connected surface. Their coefficients must therefore agree. Values at different quotient vertices are independent. The resulting basis consists of the vertex-link vectors $\ell_1, \dots, \ell_v$, each with one triangle at every corner of its vertex and no other discs. Rank-nullity gives

$$\dim V_Q \leq v + q. \tag{3}$$

In particular, every connected triangle-only normal surface is one vertex link.

Choose one representative of each component type of F having a nonzero quadrilateral coordinate. Retain a two-sided representative unchanged. Replace a one-sided representative S by the horizontal frontier of a small normal interval neighborhood. The frontier is the connected double cover associated to the nontrivial normal line bundle of S ; it is two-sided and has standard vector $2\mathbf{x}(S)$. For a surface with boundary, the vertical boundary of its interval neighborhood lies in ∂M , and the horizontal frontier is properly embedded.

All these neighborhoods can be chosen mutually disjoint and disjoint from retained representatives. Coalesce equal resulting vectors, and let r be their number. Adjoin all v vertex links, taking them sufficiently close to the triangulation vertices to avoid every chosen surface. The resulting $r + v$ surfaces are mutually disjoint, connected and two-sided. Their vectors are distinct: the first r have nonzero quadrilateral part, whereas vertex links do not. They lie in V_Q and are independent by Lemma 5.2. Thus

$$r + v \leq \dim V_Q \leq v + q, \quad r \leq q.$$

Each resulting nonvertex vector $\mathbf{z}$ has at most two original preimages: a two-sided type with vector $\mathbf{z}$, and a one-sided type with vector $\mathbf{z}/2$. Two different one-sided vectors cannot have the same double. Hence there are at most $2r \leq 2q$ original nonvertex types, and at most v triangle-only types. This proves the first bound. If all components are two-sided, the replacement map is injective and the same argument gives $H \leq v + q$. $\square$

5.1 A sharp one-tetrahedron example

The factor two accounts for a real geometric possibility: a one-sided component and its connected two-sided double can coexist. The fixture `layered_torus(1)` makes this explicit. Label the tetrahedron vertices $0, 1, 2, 3$, with face i opposite vertex i , and identify face 0 with face 1 by the permutation $(0\ 1\ 2\ 3)$; faces 2 and 3 remain boundary faces. This is a one-vertex solid torus. Use coordinate order

$$(T_0, T_1, T_2, T_3; Q_{01|23}, Q_{02|13}, Q_{03|12}).$$

Its vertex-link disc and a normal Möbius band have vectors

$$\ell = (1, 1, 1, 1; 0, 0, 0), \quad \mathbf{m} = (0, 0, 0, 0; 0, 1, 0).$$

For the single quadrilateral of $\mathbf{m}$, the corners on edges 03, 23, 12 become one vertex, while its corner on 01 is another. There are three edge cells and one face cell, so its Euler characteristic is zero. The two unpaired sides form one boundary circle. The quotient is therefore a Möbius band. Its normal frontier is an annulus with vector $2\mathbf{m}$.

Choose this band, its small frontier, and the vertex link disjointly. Their union has vector

$$\ell + 3\mathbf{m} = (1, 1, 1, 1; 0, 3, 0)$$

and exactly the three distinct component vectors $\ell, \mathbf{m}, 2\mathbf{m}$. Thus $H = 3 = v + 2q$ with $v = q = 1$. The union with vector $\ell + 2\mathbf{m}$ also attains the two-sided bound $H = 2 = v + q$. The supplied native inventory and independent Regina decomposition agree on these explicit components; the full fixture and audit record are retained in `results/normal_inventory_audit.json`.

5.2 Consequences for encoded output

Suppose the supplied coordinates have at most $B \geq 1$ bits. Every component coordinate is bounded by its source coordinate. A component multiplicity is at most the total number of normal discs, hence needs $O(B + \log(t + 1))$ bits. Since $v \leq 4t$ and $q \leq t$, the coalesced normal inventory has

$$O(t^2(B + \log(t + 1)))$$

bits. The number of components itself remains unbounded for fixed triangulation; it is their number of distinct full vectors that is linear.

AHT already establish polynomial-time recovery of component data from binary normal coordinates [1, Corollary 17]. The present bound sharpens the final output accounting and permits any polynomial-time predicate on one connected vector to be evaluated on every type. It does not enumerate candidate source surfaces. Nor does equality of the weighted sum of proposed component vectors certify decomposition: normal sums can reconnect surfaces, so actual component membership still comes from the checked orbit calculation.

6 Polynomial representation of raw singleton blocks

The existing word arena reconstructs substituted concatenations after every acyclic elimination batch. Its one-batch polynomial bound does not, by itself, give a polynomial bound for an arbitrary number of batches. We now give a different representation with a bound for an entire sequence of raw singleton eliminations. The representation has a tested reference implementation, but it is not integrated into the knot-group producer or the independent source-replay checker.

SLPs and their use in group presentations are established: for example, Theorem 4.10 of Ennes and Maria's *Compressed data structures for Heegaard splittings* (arXiv:2507.11406v2) converts compressed relators to a short ordinary presentation by introducing SLP nonterminals as Tietze generators [3]. The contribution here is an explicit representation, size recurrence and prototype for the cumulative cost of a complete raw singleton-elimination block. No claim of literature-wide priority is needed for this refinement.

6.1 Signed circuits and mutable generator bindings

Fix a finite generator alphabet X of size r_0 . A signed handle v references a circuit vertex; $-v$ represents the formal inverse word of v . Vertex zero represents the empty word. There is one terminal vertex for each positive generator, and immutable concatenation vertices $c(u, v)$. Consequently

$$\text{val}(-c(u, v)) = \text{val}(-v) \text{val}(-u).$$

Inversion requires no additional circuit vertices. Initially the circuit is a binary SLP. Later, a terminal for an eliminated generator receives one binding to a signed handle. An unbound terminal denotes its live generator, while a bound terminal denotes the current value of its target. All dependencies, including bindings, must form a directed acyclic graph. Concatenation vertices retain their original children for the whole block; only previously unbound generator terminals can acquire bindings.

The represented words are raw words. In particular, the adjacent letters xx^{-1} contribute two occurrences of x , even though their product is trivial in the free group. This restriction makes a zero occurrence count equivalent to the absence of a path to the corresponding live terminal. It is essential to the proof below.

The current presentation retains the original list of relator slots. Eliminating a generator clears exactly its chosen donor slot. Every other root remains a reference to its existing circuit, so a new generator binding applies the substitution to all retained relators simultaneously. Historical donor vertices can remain allocated. Their continued presence does not affect the presentation and is included in all size bounds.

6.2 Position-free extraction of a singleton context

Let x be live and let a donor root represent

$$W = Lx^\varepsilon R, \quad \varepsilon \in \{1, -1\},$$

with exactly one occurrence of x or x^{-1} . For this one generator, compute counts in the three-element set $\{0, 1, 2\}$, where 2 means at least two. Counts add with saturation at concatenations, pass through bindings, and are unchanged by orientation. A topological pass therefore recognizes the singleton condition exactly. Counts need no expanded-length integers, substring coordinates or word-equality oracle.

Starting at the donor root, follow its unique path to the live terminal x . At a negative concatenation, reverse the order of the children and negate their handles. At a binding, follow the bound handle with the appropriate sign. At each concatenation on the path, exactly one child contains x and the other contains none. Save this other child as a signed sibling handle. Siblings before the path form L in descent order; siblings after the path form R in reverse descent order. This yields an ordered list of existing signed handles for RL .

Combine the list by balanced pairwise concatenation. If there are q nonzero handles, at most $\max(0, q - 1)$ new concatenation vertices are needed, and every path through the new concatenations has length at most $\lceil \log_2 \max(1, q) \rceil$. The balancing is by the number of sibling handles, not by the expanded lengths of their values. Interning repeated concatenations is optional and can only reduce allocation.

Install the binding

$$x \mapsto \begin{cases} (RL)^{-1}, & \varepsilon = 1, \\ RL, & \varepsilon = -1. \end{cases} \quad (4)$$

This is the familiar exact solution of $Lx^\varepsilon R = 1$.

Lemma (acyclicity and simultaneous substitution). The binding in (4) preserves acyclicity and applies exactly the corresponding singleton Tietze elimination to all retained roots.

Proof. Before the binding is installed, every saved sibling has occurrence count zero for x . Hence its old dependency graph has no path to the live terminal x . The new image consists only of these siblings and new concatenations. It consequently has no path to x . Any newly created directed cycle would have to use the new edge out of x and then return from the image to x , which is impossible.

Let ϕ evaluate the old binding graph as raw words in the live alphabet, and let σ replace x by the raw word specified in (4). Evaluation of the enlarged acyclic graph is $\sigma \circ \phi$: this follows directly by induction

over its dependency order, with the only changed terminal being x . In particular, bindings of earlier eliminated generators update their values automatically. Deleting the donor equation and eliminating x is an elementary Tietze transformation, so the presented group is preserved. $\square$

The no-path argument remains valid even when a later image references an earlier eliminated terminal. If that earlier terminal could lead back to the current generator, its occurrence count in the proposed image would be positive and it could not be a saved sibling. No fixed elimination order on original generator labels is required.

6.3 The global size theorem

Write S_0 for the initial number of nonzero vertices and D_0 for the largest number of concatenation vertices on a dependency path in the initial SLP. Suppose that exactly $k \leq r_0$ distinct live generators are eliminated, with no intervening free reductions, new generators, or other word transformations. Donors may be chosen adaptively, but each must satisfy the raw singleton condition at the time it is used.

After i eliminations let D_i be the largest number of concatenation vertices on a path when all generator bindings are ignored, and let H_i be the longest full dependency path, counting concatenations and binding edges. At most i bindings lie on any full path: a directed acyclic path cannot visit a terminal twice. Such a path splits into at most $i + 1$ binding-free segments. Therefore

$$H_i \leq (i + 1)D_i + i < (i + 1)(D_i + 1) =: A_i. \quad (5)$$

The singleton descent encounters at most H_i concatenations, so its sibling count q_i is at most A_i . Balanced construction gives the recurrence

$$D_{i+1} \leq D_i + \lceil \log_2 \max(2, A_i) \rceil. \quad (6)$$

Theorem (complete raw singleton blocks have polynomial size). For every $0 \leq i \leq k$,

$$D_i \leq D_0 + 6i \log_2(D_0 + i + 2). \quad (7)$$

In particular, putting $T = D_0 + k + 2$, the number S_k of allocated nonzero vertices satisfies the explicit bound

$$\begin{aligned} S_k \leq S_* := S_0 + \frac{k(k+1)}{2}(D_0 + 1) \\ + 2k(k-1)(k+1) \log_2 T. \end{aligned} \quad (8)$$

The bit length B_k of every represented raw length, including historical roots and image vertices, satisfies

$$B_k \leq (k+1)(D_0 + 6k \log_2 T + 1). \quad (9)$$

Thus

$$S_k = O(S_0 + k^2(D_0 + 1) + k^3 \log(D_0 + k + 2)),$$

and a whole block has polynomial encoded complexity in the initial grammar size and rank, even if it uses a linear number of serial phases.

Proof. Set $T_i = D_0 + i + 2 \geq 2$. The claim for D_0 is immediate. Suppose that (7) holds at i . Since $i + 1 \leq T_i$, $D_0 + 1 \leq T_i$, and $i \leq T_i$, we obtain

$$A_i \leq T_i^2(1 + 6 \log_2 T_i) \leq T_i^5.$$

For the final inequality it suffices to observe that $\log_2 t \leq t - 1$ for $t \geq 2$, and $6t - 5 \leq t^3$ on the same range. The latter follows because $t^3 - 6t + 5$ is positive at $t = 2$ and has positive derivative thereafter. As $T_i^5 \geq 2$, it follows that

$$\lceil \log_2 \max(2, A_i) \rceil \leq 5 \log_2 T_i + 1 \leq 6 \log_2 T_i.$$

Substitution into (6) yields

$$D_{i+1} \leq D_0 + 6(i+1) \log_2 T_i \leq D_0 + 6(i+1) \log_2 T_{i+1},$$

completing the induction with the stated constant.

At step i , no generator vertex is added and at most $q_i \leq A_i$ concatenation vertices are allocated. Sum

$$A_i \leq (i+1)(D_0+1) + 6i(i+1)\log_2 T$$

over $i = 0, \dots, k-1$. The identities

$$\sum_{i=0}^{k-1} (i+1) = \frac{k(k+1)}{2}, \quad \sum_{i=0}^{k-1} i(i+1) = \frac{k(k-1)(k+1)}{3}$$

give (8).

Unfolding a binary/unary acyclic circuit of height H_i gives a tree with at most 2^{H_i} terminal leaves. Inversion does not change length, and bindings are unary. Hence every represented length has bit length at most $H_i + 1$. Combine this observation with (5) and (7) to obtain (9). Empty values have bit length zero and satisfy the same bound. $\square$

6.4 Running time and ordinary SLP export

For a supplied sequence of donor slots and live generator labels, the reference algorithm performs a topological pass and a capped occurrence pass over all historical vertices at each step. The graph has at most two outgoing edges per vertex. Context extraction and balanced assembly take $O(q_i)$ further operations. The whole block therefore uses $O(kS_*)$ graph operations beyond initial input handling. If M is the original number of relator slots, reading the input and exporting its roots add $O(S_0 + M)$ operations; the total, including final export, is $O(S_0 + M + kS_*)$. Working storage is $O(S_* + M)$, including the fixed relator-root list and the k binding entries. The exported root list uses $O(M \log(S_* + r_0 + 2))$ bits. Input labels may be renumbered densely once; node identifiers use $O(\log(S_* + r_0 + 2))$ bits.

This count requires neither large-integer lengths nor exact string equality. A deterministic implementation can omit interning and use dense arrays for all generator and vertex metadata. If interning is desired, balanced comparison dictionaries give a conservative $O(kS_* \log^2(S_* + r_0 + 2))$ bit-operation accounting in the usual bit-cost random-access model. The Python prototype uses hash dictionaries; its measured timings do not constitute that deterministic machine bound.

Adaptive donor discovery remains polynomial within this raw-only model: one may compute the same saturated counts for each live generator and scan every retained root, adding a factor at most r_0 to the graph passes plus the root-scanning cost. This observation gives no guarantee that another singleton donor exists, that a chosen order reaches rank one, or that searching over many alternative orders is cheap.

The representation is not a permanent barrier to existing word kernels. At an explicit end-of-block boundary, topologically evaluate each vertex in both orientations. A bound generator becomes an alias to its image; an unbound generator becomes its positive and negative terminal letters; a concatenation becomes two ordinary concatenations with opposite child orders and signs. This exports a standard binary SLP containing at most $2S_k$ nonzero vertices, with all references in allocation order and no bindings. The export is linear in circuit size. It is implemented by `export_slp` in the prototype and checked against a separate literal reconstruction on the finite tests.

A single existing polynomial compressed free-reduction operation [9] can therefore be charged after the export. The theorem does not cover a repeated alternation of normalizations and other presentation moves. Nor does it automatically include torsion-dependent primitive-power quotients whose donor generator is not a raw singleton. Those operations require their own growth analysis.

6.5 A controlled family with exponentially long raw contexts

Let the generators be $a, b, x_0, \dots, x_k, z$, and take

$$r_0 = x_0^{-1} a x_1 b, \quad r_i = x_0^{-1} x_{i+1} \quad (1 \leq i < k), \quad r_k = x_0^{-1} z.$$

Eliminate x_i from the current value of r_i , successively for $i = 0, \dots, k-1$, and retain r_k . After the first elimination, $x_0 = A_1 x_1 B_1$ with $A_1 = a$, $B_1 = b$. If after step j it is $A_j x_j B_j$, the next donor gives $x_j = A_j^{-1} x_{j+1} B_j^{-1}$, so

$$A_{j+1} = A_j A_j^{-1}, \quad B_{j+1} = B_j^{-1} B_j.$$

These are raw words, so $|A_j| = |B_j| = 2^{j-1}$ and the retained relator has length $2^k + 2$. All selected pivots remain singletons. This artificial presentation family stresses retained contexts and repeated substitutions; it is not a family of difficult knot diagrams. Free reduction would remove its deliberately retained cancellation padding.

The benchmark compares the new circuit with the pinned repository’s `apply_batch` called with the very same one-entry donor sequence. Both arms start from the same explicit source relators, clear the same slots, retain raw words, and disable their expansion method. An identical eager control measures scheduling variability. There is one excluded warm-up per size and arm, followed by five randomized A/A/B rounds. Timing includes source construction and all eliminations; final diagnostic measurements are excluded. Both arms use caps of two million vertices and one billion implementation work units, and every run completes.

k	Eager (s)	Control (s)	Persistent (s)	Eager vertices	Persistent vertices
8	0.000879	0.000869	0.000363	238	83
16	0.005813	0.005988	0.001871	914	279
32	0.043468	0.045512	0.010994	3610	839
64	0.341194	0.348591	0.067422	14378	2534
128	2.853190	3.013364	0.395421	57418	6890

The final row gives a measured median speedup of about 7.22 relative to the eager arm. Its retained word has length $2^{128} + 2$. The new representation has 128 extra binding entries in addition to its 6,890 vertices, and uses signed handles instead of separate inverse subgraphs; vertex counts are consequently structural counts, not byte measurements. Both sides retain historical vertices. Work counters have different charging conventions and should not be interpreted as portable CPU instruction counts.

The six test methods include 600 deterministic random presentation traces checked against independently implemented literal substitution, explicit negative handles and empty images, repeated-child rejection, forward bindings, interruption before a binding is published, and the 128-step capacity example with expansion disabled. The tests check the syntactic height recurrence and the $2S_k$ export bound as well. These are tests of word representation and Tietze semantics; no new knot verdict is produced.

6.6 Consequences and next questions

The size theorem removes a logarithmic-phase hypothesis from this restricted raw singleton-elimination component, provided the persistent representation is used. It does not establish a quasi-polynomial unknot-recognition algorithm. The implementation work needed before production use is concrete: add a source-derived replay path; preserve all presentation-slot and cancellation semantics; implement a compatible raw donor planner; export exactly at normalization boundaries; and measure complete discovery and replay on real diagrams. Algebraically, the next questions are whether the cubic logarithmic size term can be improved, whether suitable normalization steps preserve a comparable amortized bound, and which topologically meaningful diagram classes admit enough raw singleton eliminations to reach a checked terminal presentation.

7 Implementation, independent checks, and measurements

7.1 What changed in the maintained code

The integration patch adds three production-capable geometry modules and three test modules. It does not alter existing recognition dispatch or replace the dense incidence API. The additions are `weighted_orbits.py`, `sparse_port_incidence.py`, and `normal_components.py`. The persistent circuit remains an explicitly separate research prototype in the experiments directory, pending the planner and source-replay work described below.

The supplied patch applies to the pinned commit; the accompanying source snapshot includes the unchanged dependencies needed for independent reproduction. Historical tests import small reference implementations from four sibling report directories. Those fixtures and their original licences are included with their hashes. The first attempt to run the full suite failed at import/setup because those sibling files were absent from the sparse checkout. Restoring the unchanged pinned fixtures resolved that environment problem; the original failed log is retained rather than silently discarded.

7.2 Tests that exercise mathematical failure modes

The final maintained suite reports 1,044 passing test methods, with Regina available. Of these, 26 are the new geometry methods: 14 for weighted orbits, five for sparse incidence, and seven for component extraction. The separate persistent-circuit suite has six passing methods. These method counts understate the number of mathematical instances, because the methods contain deterministic exhaustive and randomized comparisons.

Audit	Instances	Independent comparison or property
Small weighted orbit systems	15,816	Expanded union-find orbit sums for all systems with at most two pairings and binary point weights through universe size four
Signed vector weights	2,500	Expanded signed three-coordinate orbit sums; 150 also checked with the classical merger
Full-range transfers	5,000	Literal pointwise pushforward versus compressed folding
Sparse incidence	250	Existing dense histogram versus sparse attained supports
Focused normal extraction	72	Exact Regina component-coordinate multisets
Broader normal audit	1,000	Regina components, topology, disk status, source-bound replay, and profile inequalities
Persistent elimination	600 traces	Separate literal substitution oracle; more than 1,000 successful eliminations

Table 2: Mathematical comparison workloads, in addition to malformed-input, resource, cancellation, and source-binding tests. Instance collections may overlap and are not added to obtain a misleading single total.

The weighted tests include zero-dimensional vectors, negative weights, the sharp three-run example, and analytic systems with 18,001-bit endpoints. They mutate weights, dimensions, multiplicities, source pairings, and local proof records. Verification is also exercised with the orbit search function replaced by a function that raises if called. Certificate schema validation rejects Boolean values masquerading as integers.

The broader normal audit uses eleven finite triangulations, including layered solid tori, boundary-cap extensions, an interior vertex subdivision, and a solid torus connected-summed with $\mathbb{RP}^3$. The last fixture deliberately tests geometry beyond knot exteriors. It includes 200 relabelings and 63 random compatible sums, with 150 cases containing nonorientable boundary surfaces and 26 containing closed nonorientable surfaces. There are 116 cases with compressing disks. All 1,000 component multisets agree exactly with Regina. The stronger two-sided profile bound is checked in all 824 applicable cases. Every certificate replays with both orbit and parity search disabled. These are independent finite checks, not substitutes for the proofs.

7.3 Geometry timing protocol

Measurements use Python 3.12.14 on the recorded Linux x86-64 environment, with Regina engine 7.4 and distribution 7.4.1 for the independent audits. Comparative timings run without concurrent audit or full-suite workloads. Before and after each experiment the scripts hash the relevant source files and require equality. All raw samples, shuffled orders, exact inputs, completion statuses, and structural counters are retained.

The primary geometry experiment has thirteen cases and four arms: the existing dense query, an identical dense control, the new sparse query, and the new query followed by certificate verification. Each case has one excluded warm-up per arm and five measured shuffled rounds, giving 260 measured calls. Timing covers the complete call. Histogram comparison is outside the timing interval; proof replay is inside the verified arm. No local resource cap censors these runs.

The cases distinguish disjoint, nested, and coincident ports. Static systems on 2^{512} points expose the unavoidable cost of dense output without geometric scheduling work. Native interval systems come from an eight-tetrahedron normal meridian and from 2^{64} parallel meridians. These are supplied normal surfaces, not knot-recognition benchmarks.

The median paired ratio is formed inside each shuffled round and then aggregated; it need not equal the

Case	Dense	Sparse	Sparse + replay	D/S	A/A
Static, disjoint, $r = 4$	0.451	0.126	0.201	3.06	1.05
Static, disjoint, $r = 8$	15.045	0.184	0.324	81.55	0.97
Static, disjoint, $r = 12$	477.954	0.323	0.685	1390.69	1.05
Static, nested, $r = 12$	15.368	0.241	0.496	61.42	0.98
Static, identical, $r = 12$	15.384	0.158	0.343	99.07	0.99
Meridian, disjoint, $r = 4$	12.606	2.496	3.604	5.04	1.05
Meridian, disjoint, $r = 6$	38.299	2.908	4.212	14.07	0.99
Meridian, disjoint, $r = 8$	171.075	3.063	4.735	56.36	0.99
Meridian, identical, $r = 8$	3.223	2.522	3.348	1.23	0.98
Parallel, disjoint, $r = 4$	25.029	2.913	4.364	8.66	1.00
Parallel, disjoint, $r = 6$	94.974	3.269	4.868	28.99	1.00
Parallel, disjoint, $r = 8$	382.624	3.612	5.937	107.92	1.00
Parallel, identical, $r = 8$	6.542	2.721	3.584	2.29	1.01

Table 3: Final primary geometry experiment. Times are median milliseconds per complete call. D/S and A/A are medians of paired dense/sparse and dense/control ratios. The full JSON retains all four arms and all samples.

ratio of the two displayed medians. The A/A column is the dense median paired ratio against its identical control. Values away from one expose the variability of these short measurements. Large disjoint-port gains are consistent with eliminating the dense subset enumeration. Small gains or reversals are treated more cautiously, especially where identical ports already benefit from the baseline’s marked-union reuse.

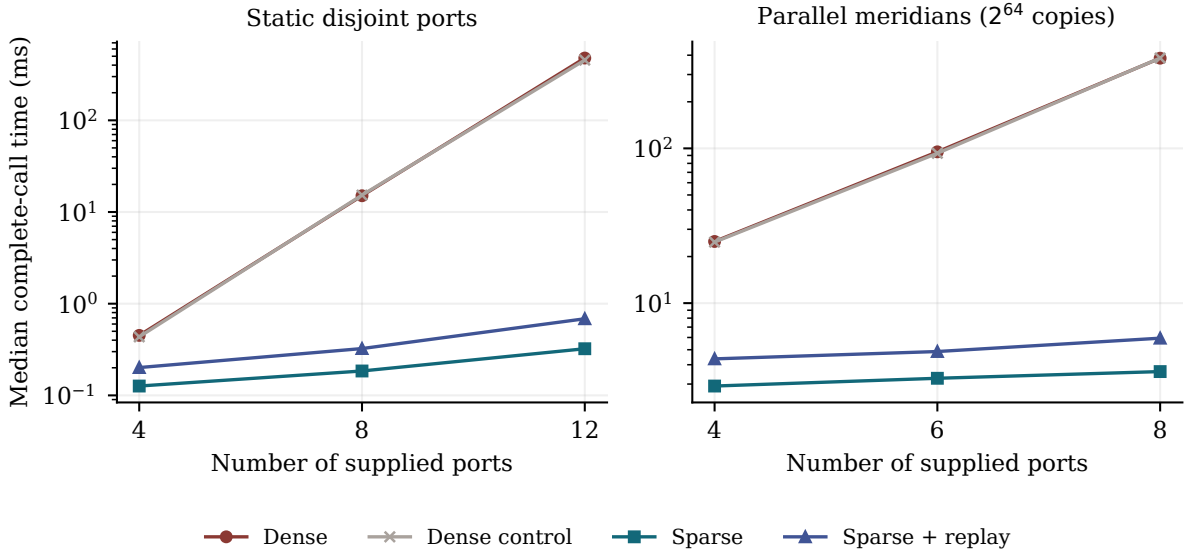


Figure 2: Completed dense and sparse port queries on the final source. The horizontal axis counts explicit ports. Static disjoint ports isolate the dense output cost; the native parallel-meridian cases also exercise the interval scheduler. The axes are logarithmic where indicated. Points are medians, not fitted complexity exponents.

7.4 Batched follow-up and compressed capacity

The initial measurement pass showed material A/A variation on small queries. The final primary pass and a separate follow-up are both retained; the original pass is labelled as a development snapshot because minor cancellation-polling corrections followed it. The follow-up uses repeated complete calls within each timed batch for four predetermined small or noisy cases, with seven shuffled rounds. It reports time per call and does not replace the primary experiment selectively.

Case	Batch	Dense	Sparse	Verified	D/S	D/V	A/A
Static, disjoint, $r = 4$	128	0.324	0.049	0.117	6.474	2.715	0.995
Static, nested, $r = 12$	8	14.763	0.179	0.427	81.711	33.724	0.988
Meridian, identical, $r = 8$	16	2.811	2.087	2.974	1.404	0.945	1.007
Parallel, identical, $r = 8$	16	5.793	2.481	3.544	2.324	1.621	1.002

Table 4: Batched follow-up: median milliseconds per call; seven paired rounds, 4,704 measured calls and 672 excluded warm-up calls. D/V compares dense counting with sparse counting plus proof replay. The 0.945 ratio is a retained regression.

Capacity records are separate from comparative timing. They include the mixed normal surfaces in (2) and static sparse systems with 32, 128, and 512 disjoint ports. The latter return respectively 32, 128, and 512 attained signatures. There is no attempt to allocate a dense 2^{512} -entry output, and no claimed measured speedup against such an uncompleted allocation.

b in $K = 2^b$	Produce + replay (ms)	Certificate bytes	Profiles
128	49.24	37,072	2
1024	48.51	118,337	2
4096	54.20	334,975	2
16384	77.06	1,259,441	2

Table 5: Single completed capacity probes for $Kd + (K + 1)\ell$ in an eight-tetrahedron solid torus. These are individual capacity observations, not repeated timing estimates or comparative speedup claims.

For the eight-tetrahedron mixed surfaces, the weighted trace has the same 275 explicit events, 23 scheduler cycles, and 22 transfers across these bit lengths. It uses 40 anchor intervals, dimension 56, and 35 initial and peak weight runs. The benchmark demonstrates that enormous multiplicities do not force enumeration. It also shows the real encoded cost: the largest certificate contains 1,259,441 bytes in the stated JSON format. Hexadecimal transport avoids changing Python’s process-wide decimal conversion limit. Compression of the distribution archive is separate from the certificate’s mathematical bit-size bound.

7.5 What the word measurements establish

The persistent-elimination section gives all five rows of the A/A/B comparison and describes the exact artificial source family. At 128 eliminations the new prototype takes about 0.395 seconds versus 2.853 seconds for eager serial elimination, a ratio of median times of about 7.22. Both complete, use the same supplied donors, and keep expansion disabled. The retained raw relator has length $2^{128} + 2$.

This family was chosen to exercise repeated retained contexts. Its freely cancelling padding means that it would be inappropriate to present it as a difficult family for the full knot algorithm. The experiment supports the practical value of this particular representation and the structural bound; it does not establish an exponential lower bound on the old arena, whose measured vertex growth on this family is quadratic. A future integration benchmark must include donor discovery, normalization, source replay, and actual knot diagrams.

8 Certificate contracts and integration design

8.1 The weighted proof interface

The weighted certificate contains a version, a dimension, the canonical complete weight partition (including zero runs), ordered output classes, and the source-bound ordinary orbit certificate. A class is serialized as `[vector,multiplicity]`; it is not one object per connected component. The verifier reconstructs the claimed weights from the explicit source arguments, checks the ordinary orbit proof, and replays every

weight transfer and static emission, comparing the exact ordered classes with the claim. The producer and standalone replay utility also check total orbit multiplicity and the aggregate vector explicitly.

The ordinary orbit checker is independently implemented relative to the scheduler. Weighted production and weighted verification share the small transport arithmetic, as does normal geometry reconstruction. Their independent validation is supplied by the expanded finite oracles and the Regina component comparisons. This separation should remain visible in documentation and future proof audits: invoking a shared function twice does not create an independent arithmetic implementation.

The sparse certificate additionally binds the normalized input ports and the attained histogram. The normal-component certificate binds the triangulation and original vector, the weighted proof, the exact profiles, and the independently checked boundary parity records. For a disk profile the verifier derives its boundary graph from the source geometry and the extracted coordinates. It does not trust a supplied assertion of connectedness, Euler characteristic, or essentiality.

8.2 Input provenance and the meaning of a negative result

The normal inventory's mathematical statement is: these are the connected components of this supplied normal surface in this supplied triangulation. A compressing disk is therefore a witness about that manifold. Relating it to an original knot diagram additionally requires a checked conversion from the diagram to its exterior and checked transport through all simplifications. The new modules do not silently inherit that provenance from a filename, hash, or external-engine claim.

Similarly, an inventory with no compressing disk proves only that this particular supplied surface has no such component. It does not prove that the manifold has incompressible boundary. The empty vector gives an empty inventory, not evidence of knottedness. A future complete procedure needs a justified way to choose surfaces or certify a hierarchy. The same logical distinction applies to the word prototype: exhausting all raw singleton donors is an algebraic stopping condition, not a proof that the knot is nontrivial.

8.3 Resource and cancellation semantics

The cycle allowance is charged by ordinary orbit discovery. The weighted producer currently first builds and checks that ordinary trace, then applies its weight-run, output-record, and event allowances during weighted replay. Thus these limits do not bound the earlier trace allocation. Standalone weighted replay and certificate verification can reject an overlong explicit trace before running ordinary replay, but they still receive an already materialized input object.

These are cooperative work and output allowances. They are not promises about process memory, JSON parser allocation, or wall-clock latency. Callbacks are polled in the source and certificate loops; callback exceptions propagate instead of being mistaken for malformed data. An incomplete weighted result has no count, class list, aggregate, or certificate, and the higher-level adapters return an inconclusive status. The implementation never presents a partial inventory as complete.

For the persistent circuit, a failed or interrupted construction may allocate unreachable temporary concatenations before publishing a binding. The mathematical size theorem counts a completed valid block from a fixed initial circuit; repeated failed attempts that leave orphan allocations are not free. A production transactional wrapper should roll back those allocations, collect unreachable nodes, or include their costs explicitly in the potential function. The current tests check that an interrupted operation does not publish a false generator binding.

8.4 A concrete integration sequence

The geometry patch can be reviewed and applied without changing the recognizer's existing decisions. Its first callers should be offline normal-surface and boundary-pattern experiments. Such callers can reuse a single inventory for disk selection and support queries. If they truly need a dense table, an explicit final conversion preserves that interface and acknowledges its output cost.

The persistent circuit needs more work before production use. Its mutable generator bindings change the values of previously allocated nodes. Existing caches in the immutable word arena cannot be reused unchanged. Occurrence counts, lengths, Whitehead summaries, and donor frontiers need a dependency-aware invalidation policy or an explicit recomputation phase. The proposed first integration should keep a raw-only block, recompute saturated donor counts, and export once at the normalization boundary.

For independent replay, each pivot image must be derived from the source-bound donor and its unique raw occurrence. A certificate may specify the donor slot and live generator, but it must not be allowed to supply an unverified replacement image. The checker can independently reconstruct the signed circuit and compare the final exported presentation with the existing terminal certificate contract. Replacing a search arena alone does not establish this proof path.

The final acceptance experiment should compare the complete integrated pipeline with identical resource conventions, retain unchanged-case controls, report every timeout as censored, and include native hard-knot and hard-unknot families. That experiment is intentionally a future milestone: the current package contains kernels and a representation prototype, with their exact scope documented.

9 Cost composition and the next research obligations

The results above improve two representation costs: extracting the components of a supplied normal surface, and retaining a complete block of raw singleton substitutions. The next theorem states precisely how such local improvements could contribute to a complete recognition bound. It also makes clear which quantities future work must control.

9.1 A conditional theorem for the whole computation

Let $n \geq 2$ be the bit length of a diagram encoding. Standard encodings whose size is polynomial in the crossing count give the same quasi-polynomial complexity class. All subsequent sizes are measured in bits, including integer coordinates, combinatorial incidence, word circuits, retained history and proof references.

Proposition 9.1 (Conditional cost composition). *Suppose a deterministic algorithm constructs and evaluates a search tree for an input of size n , subject to the following uniform bounds. The transition and terminal rules are sound, and exhausting the tree is complete for both unknot and knotted verdicts. The tree has depth at most $h = h(n)$, and each node at depth $j < h$ has at most $b_j = b_j(n) \geq 1$ children. Failed candidate attempts and repair iterations are included in the tree or in its explicitly charged node work.*

Every intermediate state has at most $s = s(n) \geq 1$ bits. All work at one node, including generation of its candidates and rejected proposals, has bit cost polynomial in $n + s$, with one polynomial independent of the input and node. Each node contributes at most polynomially many bits in $n + s$ to a source-bound proof record. The assembled proof is independently checkable in time polynomial in its full length and n .

Then there is an absolute constant a such that discovery, proof output, and independent verification together have cost at most

$$\left((h+1)(n+s+2) \prod_{j=0}^{h-1} b_j \right)^a. \quad (10)$$

In particular, for a fixed $c \geq 1$, the sufficient condition

$$\log_2(n+s+2) + \log_2(h+1) + \sum_{j=0}^{h-1} \log_2 b_j = O((\log n)^c) \quad (11)$$

gives a complete deterministic recognition algorithm of complexity $2^{O((\log n)^c)}$.

Proof. The number of nodes at depth j is at most $\prod_{i=0}^{j-1} b_i$. Since every $b_i \geq 1$, the total node count is at most $V = (h+1) \prod_{i=0}^{h-1} b_i$. Summing the uniform node cost and proof length gives $V(n+s+2)^{O(1)}$ for each. Applying the assumed polynomial verifier to that full proof preserves a fixed polynomial in V and $n+s+2$, proving (10). Taking logarithms gives (11). Soundness and completeness, which are separate hypotheses, turn this computation bound into a recognition bound. $\square$

For example, $s = n^{O(1)}$, $b_j = n^{O(1)}$ uniformly, and $h = O(\log n)$ give $2^{O((\log n)^2)}$. Replacing the depth by $O((\log n)^2)$ gives $2^{O((\log n)^3)}$. A deterministic chain with polynomially many steps and polynomial state size needs no logarithmic-depth hypothesis: its branching factors are all one. A small description of each state does not establish the branching term; polynomial-time operations do not establish a bound on the largest state. If the terminal rules only certify unknottedness, the same analysis bounds a possibly inconclusive positive search and does not supply the missing negative verdict.

The analogous accounting for an iteration bound $L(g+1)^L$, as in Lackenby’s hierarchy analysis [6, Proposition 9.1], is

$$\log_2(n+s+2) + \log_2(L+1) + L \log_2(g+1) = O((\log n)^c).$$

This is conditional on bounded candidate generation, repairs and replay throughout the computation. Bounds for an essential boundary pattern cannot simply be applied to arbitrary unsuccessful branches.

9.2 Prioritized questions and concrete targets

The following questions are ordered by their role in completing the geometric and algebraic pipelines. Each asks for a specific constructive result or obstruction, rather than an improvement to isolated timings.

1. **How can a useful surface be found?** Can one generate a complete family of useful normal or hierarchy surfaces after $2^{\text{polylog } n}$ candidate attempts, with uniformly bounded encoded coordinates? Component extraction now permits a candidate to be disconnected. The missing statement concerns coverage: whenever the required compression or hierarchy move exists, some generated candidate must contain it. A first result should specify an actual diagram or triangulation class and a parameter controlling this search, including the cost of finding that parameter’s witnessing structure.
2. **Can cutting be performed directly on compressed coordinates?** Given a validated triangulation and a binary normal vector, construct a compressed cell or handle description of the cut manifold, together with its boundary pattern and a checkable identification with the geometric cut. The target is polynomial cost in the input and output bit lengths, without allocating one cell per normal disc. Weighted segment tracking under crushing already has a concrete precedent in the work of He, Sedgwick and Spreer [4, Section 4.1]; a full checked cut representation requires additional topology and attachment data. State exactly which parts require essentiality hypotheses and provide a defined outcome when a candidate does not satisfy them.
3. **Which ordered attachment data admit small descriptions?** Extend sparse component incidence to boundary order and transport of named arcs, points and intervals. An acceptable representation must support the actual gluing operations, not merely count incidences. Determine whether a useful class has polynomial-size attachment maps, and construct lower-bound examples outside it. The linear bound on unmarked component vectors does not bound the number of possible markings; relative offsets alone can distinguish many states.
4. **How is peripheral provenance maintained?** Connect the input diagram to a finite exterior triangulation with checked meridian and longitude data. Then carry that data through cutting, crushing, retriangulation and any change of combinatorial coordinates. The desired output is a composable source-bound certificate for the peripheral identifications. Recognizing an essential disc in a supplied torus-boundary manifold is insufficient until that manifold and its markings are linked to the original diagram.
5. **How small can a selected-component witness be?** The complete coalesced inventory is polynomial and has at most $v + 2q$ types. Can a particular disc or marked component be certified with a substantially smaller record than the whole weighted trace, while still proving connectedness and source membership? Investigate persistent anchors, locally checkable transport summaries, and streaming replay. Any savings must include locating the component and verifying its boundary class; conservation of the total vector is not a substitute.
6. **Can the geometric checker have a smaller trusted core?** Separate combinatorial manifold validation, interval equivalence, integer weight transport and surface classification into explicit proof interfaces. Formalize the local transport and normal-anchor statements, or implement an independently derived second checker. Measure proof bytes and maximum replay state as well as arithmetic work. Resource limits must preserve the distinction between an invalid certificate and an unfinished computation, and malformed records must be charged by their actual encoded length.
7. **Can persistent elimination be integrated without changing semantics?** Add source-derived replay for immutable concatenations and generator bindings, retaining exact relator-slot identities and the raw singleton condition. Produce an end-to-end diagram experiment in which discovery and independent replay both use this representation. Ordinary SLP export provides a clear boundary to other word kernels. The mathematical target is an integrated polynomial bound for each complete

raw block, including donor identification and proof construction, rather than the substitution operation alone.

8. **What controls growth across normalization boundaries?** A polynomial bound for one raw block does not automatically survive repeated free reduction, cyclic normalization and reconstruction. Seek an amortized potential controlling all allocated grammar vertices, integer metadata and proof records across the entire sequence. A useful theorem should bound the total composition, not repeatedly apply a fixed polynomial whose degree compounds with the number of boundaries. Counterexamples for particular normalization policies would identify which kinds of sharing must be retained.
9. **When does donor planning admit a complete efficient policy?** Computing all current raw singleton counts is polynomial, but choosing an order that reaches a terminal presentation may involve many choices. Identify an exchange property, a monotone dependency structure, or a restricted-width class on which a specified policy is complete. Charge failed exposure moves and alternative orders explicitly. A promising first theorem would connect such an algebraic class to a recognizable family of diagrams, rather than assuming a successful donor sequence as additional input.
10. **Can hierarchy repairs be accelerated with a proved potential?** Construct the actual correction operations and bound both their number and the sizes of the descriptions they create. A logarithmic-depth theorem needs a quantity that decreases under the accelerated procedure, including unsuccessful choices and subsequent repairs. Determine whether the potential should combine handle complexity, boundary-pattern complexity and compressed attachment size. A terminating integer potential whose initial value is exponential does not meet this target.
11. **How do the geometric and algebraic branching costs compose?** Establish bounds for the sum of logarithmic branching factors in (11), allowing depth-dependent choices and shared states. Test whether reductions in one representation preserve the parameters that make the other efficient. A restricted theorem may be valuable even without a universal result, provided its admission conditions are recognizable and remain valid after every permitted transition. Switching between individually efficient backends is not itself a bound on the resulting search.
12. **Which evidence best tests the proposed global bounds?** Build reproducible diagram families that vary coordinate bit length, distinct component types, attachment complexity, raw elimination depth and normalization frequency independently. Record candidate failures, conversion, proof generation, proof bytes, replay and fallback. Include cases designed to saturate proved bounds and cases that defeat favorable heuristics. Such experiments should guide the next structural conjecture and identify its weakest hypothesis; a successful finite corpus cannot replace the required completeness and worst-case arguments.

A Reproduction and artifact map

The archive contains the article source, compiled PDF, integration patch, source snapshot, exact new tests, benchmark and audit drivers, raw JSON, build and test logs, and a SHA-256 manifest. The top-level README records the pinned commit and the commands below in their executable directory layout. Python’s standard library suffices for the weighted and word kernels; Regina is needed for the independent normal-surface audit and the optional maintained native tests. The recorded environment uses Regina distribution 7.4.1.

```
python -m unittest discover -s code/fast/tests -v
python -m unittest discover -s experiments -p 'test_persistent_elimination.py' -v
python experiments/audit_normal_inventory.py --fast code/fast --cases 1000 --output results/
normal_inventory_audit_reproduced.json
python experiments/benchmark_geometry.py --fast code/fast
python experiments/benchmark_geometry_batched.py --fast code/fast
python experiments/benchmark_persistent_elimination.py --baseline-fast code/fast --sizes 8 16
32 64 128 --rounds 5
```

The first command is run with `code/fast` on `PYTHONPATH`, or from that directory with `-s tests`; the supplied reproduction script sets the path explicitly. Run timing experiments sequentially, apart from

tests and plotting. The artifact README gives the output-path options and does not require overwriting the frozen measurement files.

The geometry JSON files use exact hexadecimal strings for exceptionally large integers; the supplied `integer_codec` accepts these without changing global interpreter settings. A standalone verifier can load the source fixture and call `verify_normal_component_certificate` with the saved object. The source hashes bind the empirical report to the files actually timed. They are reproducibility metadata, not mathematical substitutes for the replay checks.

Archive path	Purpose
<code>article/</code>	Main TeX, proof sections, figures, generated tables, bibliography, and compiled article
<code>code/fast/</code>	Maintained source snapshot plus the six new geometry files
<code>code/reports/</code>	Minimal unchanged reference fixtures required by historical tests
<code>experiments/</code>	Persistent prototype, exact tests, geometry audits, timing drivers, and result-table generation
<code>results/</code>	Full samples, inputs, proof records, hashes, logs, and explicitly labelled development snapshots
<code>integration.patch</code>	Six additive files relative to the pinned ProveIt baseline
<code>SOURCE_MANIFEST.json</code>	File-level origin and SHA-256 information

Table 6: The package is intended to be reviewable before repository integration.

B A compact review checklist for the mathematical claims

Review of the weighted construction should focus on the local invariant: every original weight contribution stays attached to exactly one current representative of its original orbit. Full-range transfer, static emission, and truncation are the operations that change this attachment. Global conservation alone is weaker and must not become the proof obligation.

Review of the profile theorem should preserve the distinction between the number of connected components and the number of distinct standard vectors. Two-sided disjoint representatives give linear independence; one-sided representatives enter through their connected normal frontiers. The dimension estimate uses actual manifold vertex links. Applying it to a different coordinate system or an ideal triangulation requires a new kernel calculation.

Review of the persistent theorem should check raw occurrence semantics, acyclicity after a binding, the binding-free depth recurrence, and the accounting of historical nodes and original relator slots. Normalization and successful singleton elimination are different operations; adding the former between every step changes the hypotheses.

Finally, every proposed general complexity conclusion should display an explicit upper bound on total explored states and their maximum encoded size. Polynomial local arithmetic, a short successful certificate, or good benchmark scaling can each be valuable without supplying those two global bounds.

References

- [1] Ian Agol, Joel Hass, and William Thurston. The computational complexity of knot genus and spanning area. *Transactions of the American Mathematical Society*, 358(9):3821–3850, 2006. Section 6, especially Lemma 15, Theorem 16 and Corollary 17. <https://doi.org/10.1090/S0002-9947-05-03919-X>; <https://arxiv.org/abs/math/0205057>.
- [2] Benjamin A. Burton. Converting between quadrilateral and standard solution sets in normal surface theory. *Algebraic & Geometric Topology*, 9:2121–2174, 2009. <https://doi.org/10.2140/agt.2009.9.2121>; <https://arxiv.org/abs/0901.2629>.
- [3] Henrique Ennes and Clément Maria. Compressed data structures for Heegaard splittings. arXiv:2507.11406v2, 31 December 2025, 2025. Theorem 4.10. <https://arxiv.org/abs/2507.11406v2>.
- [4] Alexander He, Eric Sedgwick, and Jonathan Spreer. A practical algorithm for knot factorisation. In *41st International Symposium on Computational Geometry (SoCG 2025)*, volume 332 of *Leibniz International Proceedings in Informatics*, pages 55:1–55:15, 2025. Section 4.1, tracking segments by weighted orbits. <https://doi.org/10.4230/LIPICs.SoCG.2025.55>.
- [5] Simon A. King. Almost normal Heegaard surfaces. arXiv:math/0303377v3, 2003. Theorem 2 and its historical discussion. <https://arxiv.org/abs/math/0303377>.
- [6] Marc Lackenby. Incompressible surfaces, hierarchies and unknot recognition. arXiv:2607.23350v1, 25 July 2026, 2026. Sections 9–14. <https://arxiv.org/abs/2607.23350v1>.
- [7] Mathematical Institute, University of Oxford. Marc Lackenby announces a new unknot recognition algorithm that runs in quasi-polynomial time. Announcement of Marc Lackenby’s quasi-polynomial algorithm, 2021. <https://www.maths.ox.ac.uk/node/38304>. Related author slides: <https://people.maths.ox.ac.uk/lackenby/quasipolynomial-talk-oxford.pdf>.
- [8] Vladimir Reshetnikov. ProveIt: Topology/UnknotRecognition. Git repository; pinned revision identified on the title page, 2026. Source snapshot inspected 9 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/9feb4346b4d050b305f825f0f7257f495960822a/Topology/UnknotRecognition>.
- [9] Saul Schleimer. Polynomial-time word problems. *Commentarii Mathematici Helvetici*, 83(4):741–765, 2008. <https://doi.org/10.4171/CMH/142>; <https://arxiv.org/abs/math/0608563>.